A Reproducible Pipeline for Measuring Satellite Velocity Anisotropy in IllustrisTNG-100

Your Actual Name Here

September 9, 2026

Abstract

We present a reproducible pipeline for measuring satellite galaxy velocity anisotropy $\beta(r)$ in IllustrisTNG-100 (snapshot 99, $z = 0$), validated on synthetic halos with known input β before application to real data. Using bootstrap errors that resample host halos rather than individual satellites, we measure $\beta(r)$ across five host-halo mass bins, finding a monotonic trend from tangentially-biased orbits at group scale to radially-biased orbits at cluster scale, consistent with observational and N-body expectations. We further examine environment, star-forming/quenched status, and a resolved-galaxy stellar-mass cut, and perform two resolution checks (halo mass function, stellar-mass floor) confirming our sample is unaffected by simulation resolution limits. This is an exploratory, validated pipeline and pilot analysis, not a claim of novel physics; it is intended as a baseline for future population-level and cross-simulation comparisons.

1 Goal

Satellite galaxy velocity anisotropy, $\beta(r) = 1 - \sigma_t^2(r)/(2\sigma_r^2(r))$, characterizes the shape of satellite orbits within dark matter halos: $\beta < 0$ indicates tangentially-biased orbits, $\beta > 0$ radially-biased orbits, and $\beta = 0$ isotropy. This is well-studied in the literature (e.g. Wojtak, Hansen & Hjorth 2013, hereafter WHH13) and in standard N-body simulations. The goal of this project is not to discover new physics, but to build and validate a reproducible measurement pipeline for $\beta(r)$ in IllustrisTNG-100, and to use it to produce a population-level atlas across halo mass, environment, and galaxy star-formation status. This pipeline and the resulting atlas are intended as a validated baseline in standard cosmology, against which future work testing alternative dynamical frameworks could be compared – no such comparison is attempted here, since IllustrisTNG assumes standard gravity throughout.

2 Data

We use IllustrisTNG-100 (Nelson et al. 2019), snapshot 99 ($z = 0$), accessed via the public TNG API and the `illustris_python` package. Central host halos were selected in five mass bins (Table 1), based on Group_M_Crit200, requiring at least 6 subhalos per host. Satellites were identified via `SubhaloGrNr`, restricted to within R_{200} of the host, with positions and velocities computed relative to the host center under periodic boundary conditions.

As a resolution check, we computed the halo mass function $dn/d\log_{10} M$ for the full snapshot (Figure 1). The function turns over near $\log_{10}(M_{200}) \approx -1.6$, consistent with the ∼50-particle resolution limit computed from the simulation's dark matter particle mass ($m_{\rm DM} = 5.06\times10^6\ M_\odot/h$,

Bin	$\log_{10}(M_{200}/10^{10}\,M_\odot/h)$	M_{200} range	$N_{\rm halos}$
0	[1.0, 1.3)	10^{11}–$2\times10^{11}\,M_\odot/h$	3878
1	[1.3, 1.7)	2×10^{11}–5×10^{11}	3205
2	[1.7, 2.2)	5×10^{11}–1.6×10^{12}	1754
3	[2.2, 3.0)	1.6×10^{12}–10^{13}	645
4	[3.0, 4.5)	$>10^{13}$ (cluster-scale)	127

Table 1: Host halo mass bins and sample sizes.

sourced from the TNG API). Our analysis sample begins at $\log_{10}(M_{200}) = 1.0$, roughly 2.6 dex above this limit, so halo-finding incompleteness does not affect the mass bins used here.

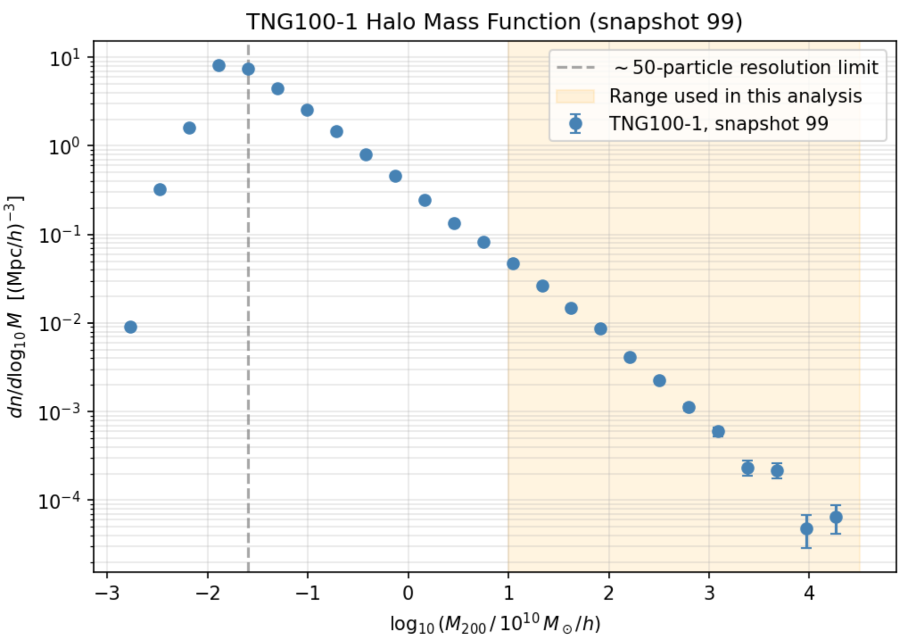

Figure 1: Halo mass function for TNG100-1 snapshot 99. The dashed line marks the ~50-particle resolution limit; the shaded region shows the mass range used in this analysis, which lies well above it.

3 Methods

3.1 Kinematic decomposition

For each satellite, position and velocity relative to the host center are decomposed into radial and tangential components. Tangential dispersion is computed as the second moment $\langle v_t^2\rangle$, not the variance of v_t: because tangential speed follows a Rayleigh distribution, using the variance introduces a bias of $+0.785(1-\beta)$, which we identified and corrected during toy-model validation. Radial dispersion uses the variance (with ddof = 1) around the bin mean, correcting for bulk flows. Errors are propagated via the delta method.

3.2 Validation on synthetic halos

Before use on real data, the pipeline was validated on synthetic halos with known input β. Across 30 random realizations each for $\beta_{\mathrm{input}} \in \{-0.5, 0, 0.5\}$, recovered values matched inputs to within $\sim 1\sigma$. A radius-dependent test case, $\beta(r) = 0.5(r/r_{\max})^2$, was also correctly recovered in shape. (See `notebooks/01_toy_validation.ipynb` for the full validation figures.)

3.3 Stacking and bootstrap errors

Satellites are stacked across all host halos within a mass bin, with radius scaled by each host's R_{200}. Radial bins use equal-count (quantile-based) edges. Uncertainties are estimated via bootstrap resampling of *host halos* (200 iterations), not individual satellites, since satellites within a single halo are not independent draws and satellite-level bootstrapping would understate the true uncertainty.

3.4 Environment and population splits

Environment is measured as the number of neighboring massive halos ($\log_{10}(M_{200}) \geq 1.0$) within a 5 Mpc/h periodic search radius. Star-forming/quenched classification uses a specific star formation rate threshold of $10^{-11}\,\mathrm{yr}^{-1}$ (Franx et al. 2008).

4 Validation

Figure 1 shows the halo mass function resolution check described above. Figure 2 shows quenched fraction as a function of clustercentric radius in the two galaxy-resolved bins (Bin 3, $N = 302$ halos; Bin 4/clusters, $N = 127$ halos), declining monotonically from 65% to 31% (Bin 3) and 94% to 70% (Bin 4) from center to outskirts. This trend is well-established in the galaxy evolution literature (ram-pressure stripping and tidal quenching are most efficient near cluster cores); we do not claim it as novel, but present it as an independent, observable-space validation that the pipeline reproduces known physics.

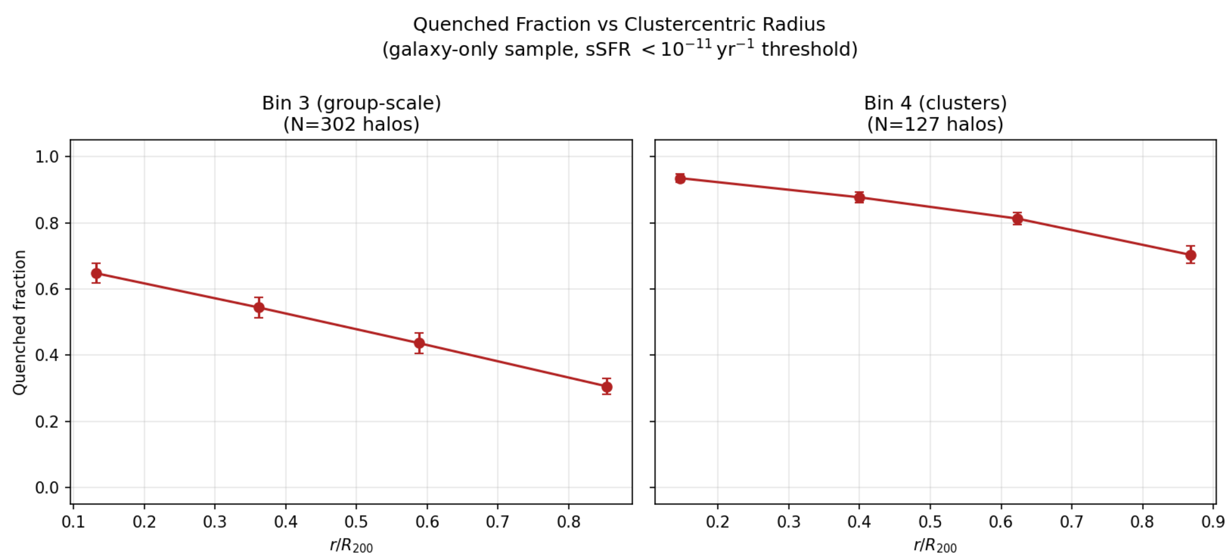

Figure 2: Quenched fraction vs. clustercentric radius for the galaxy-resolved Bin 3 and Bin 4 samples, with bootstrap errors resampling host halos.

5 Results

5.1 Mass trend

The core result (Figure 3) is a monotonic trend from tangentially-biased orbits at group scale to radially-biased orbits at cluster scale: Bin 0, $\beta \approx -0.32$ to -0.03; Bin 1, ≈ -0.18 to -0.05; Bin 2, ≈ -0.18 to -0.03; Bin 3, ≈ -0.06 to 0.02 (roughly isotropic); Bin 4 (clusters), $\approx +0.07$ to $+0.24$ (radially biased). This is consistent with low-mass halos being dynamically relaxed with tidally-circularized orbits, and cluster halos actively assembling via radial filamentary infall – matching the qualitative expectation from WHH13 and standard N-body halo anisotropy profiles ($\beta \approx 0$ at the center, rising to 0.4–0.6 in the outskirts).

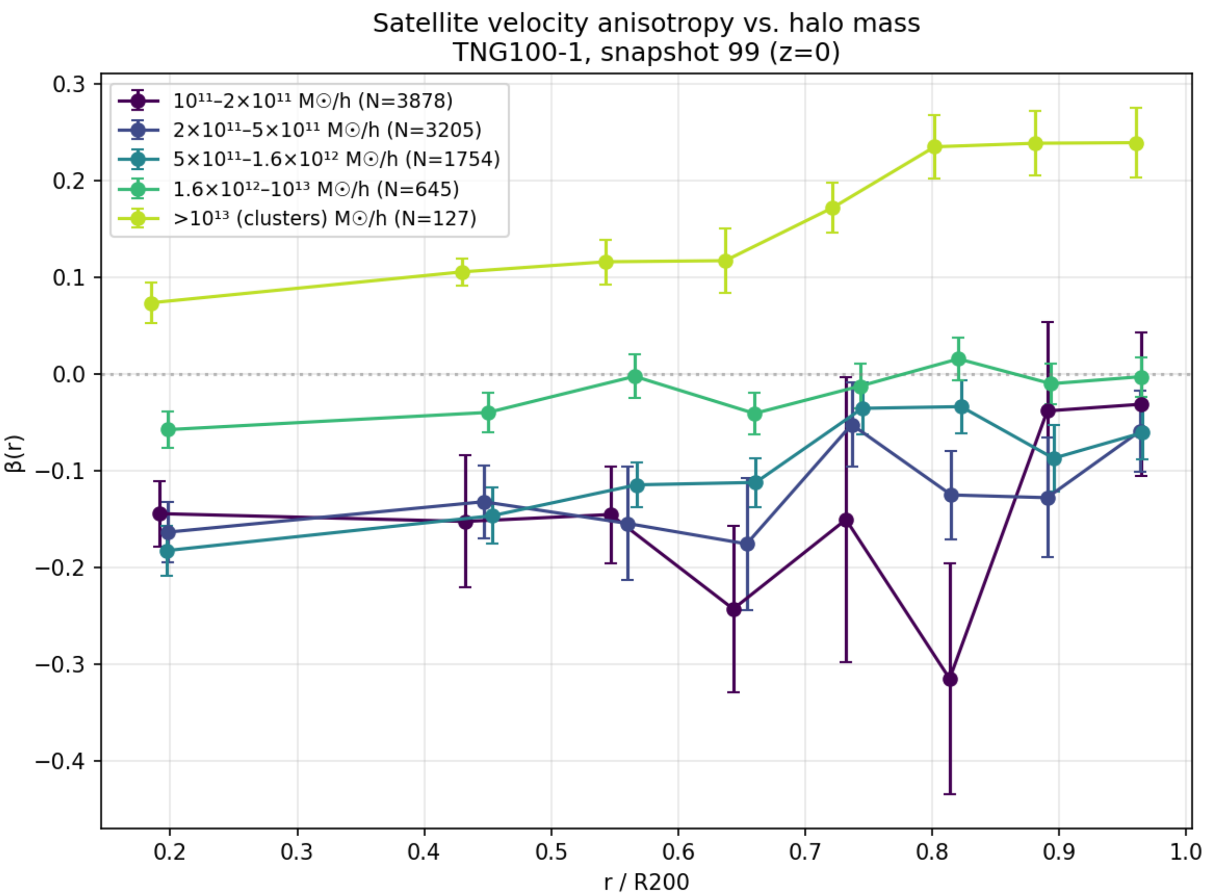

Figure 3: Stacked $\beta(r)$ profiles across the five host-halo mass bins, with bootstrap errors resampling host halos.

5.2 Environment

Per-halo Spearman correlations ($N = 2580$ individual halos with valid β) give $\rho(\beta, \text{mass}) = 0.139$ ($p = 1.55 \times 10^{-12}$). Controlling for mass via partial Spearman correlation:

$$\rho(\beta, \text{environment} \mid \text{mass}) = -0.099, \qquad p = 4.19 \times 10^{-7}, \tag{1}$$

indicating environment has a statistically significant effect on β independent of mass.

A median environment split within each mass bin shows a sign reversal in $\Delta\beta$ in $\Delta\beta$ (Figure 4): at group/intermediate mass, denser environments correspond to *more* tangential bias ($\Delta\beta$ from -0.073 to -0.084 across Bins 0–3), while at cluster mass, denser environments correspond to *more* radial bias ($\Delta\beta = +0.048$ in Bin 4). We interpret this as increased tidal stirring at group scale versus enhanced filamentary infall feeding actively-growing clusters, but note Bin 4's environment split has only 62–65 halos per half; we present this as a suggestive, physically-motivated finding warranting further investigation with larger-volume simulations, not a confirmed result.

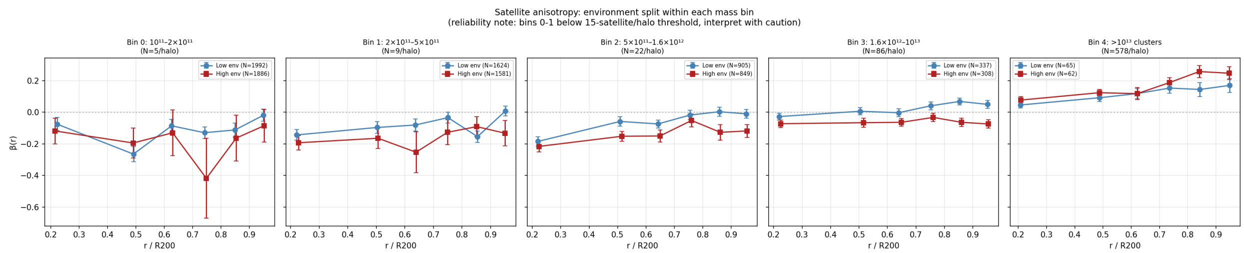

Figure 4: Median environment split of β within each mass bin. Note the sign reversal in the cluster-mass bin.

5.3 Resolution limit and galaxy-only cross-check

The vast majority of subhalos in our catalog are dark-matter-only, with zero resolved stellar mass (95.2%–99.4% depending on mass bin) – a known TNG100 resolution limit for dwarf galaxies, not a pipeline bug. Applying a resolved-galaxy stellar mass floor of $10^{8.5}\ M_{\odot}/h$, only Bins 2–4 retain usable samples (20, 302, and 127 halos respectively; Bins 0–1 have zero halos with ≥ 3 resolved satellites). Galaxy-only satellites are systematically more radially biased than the full subhalo population (Figure 5; e.g. Bin 3: $-0.019 \rightarrow +0.144$; Bin 4: $+0.162 \rightarrow +0.278$), consistent with stronger dynamical friction acting on more massive, luminous satellites.

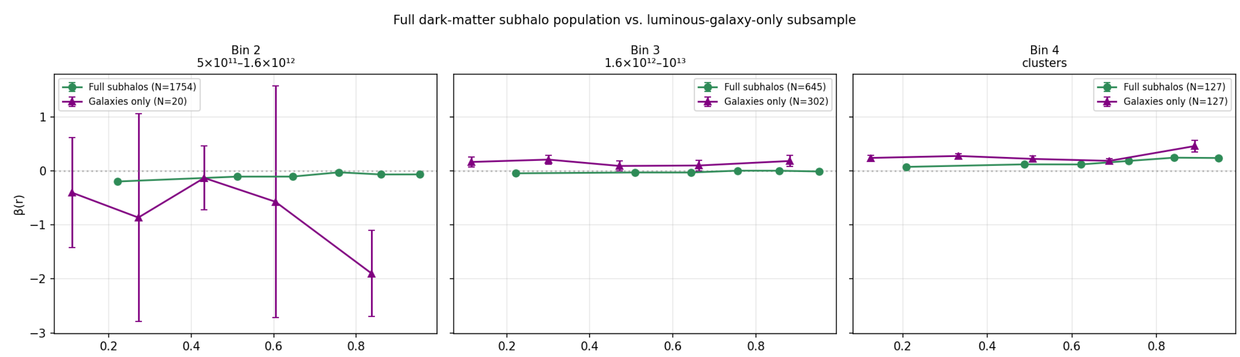

Figure 5: Comparison of β from the full subhalo population versus the resolved-galaxy-only subsample, Bins 2–4.

Restricting to the same resolved-galaxy bins, satellites also show a clear split by star-formation status (Figure 6): quenched satellites are consistently more radially biased than star-forming ones, consistent with satellites on radial, eccentric orbits plunging closer to the host center, where ram-pressure stripping and tidal quenching are strongest. Bin 2 has zero quenched-hosting halos, a physically sensible null result given group-mass halos generally lack the hot gas atmosphere needed for ram-pressure quenching.

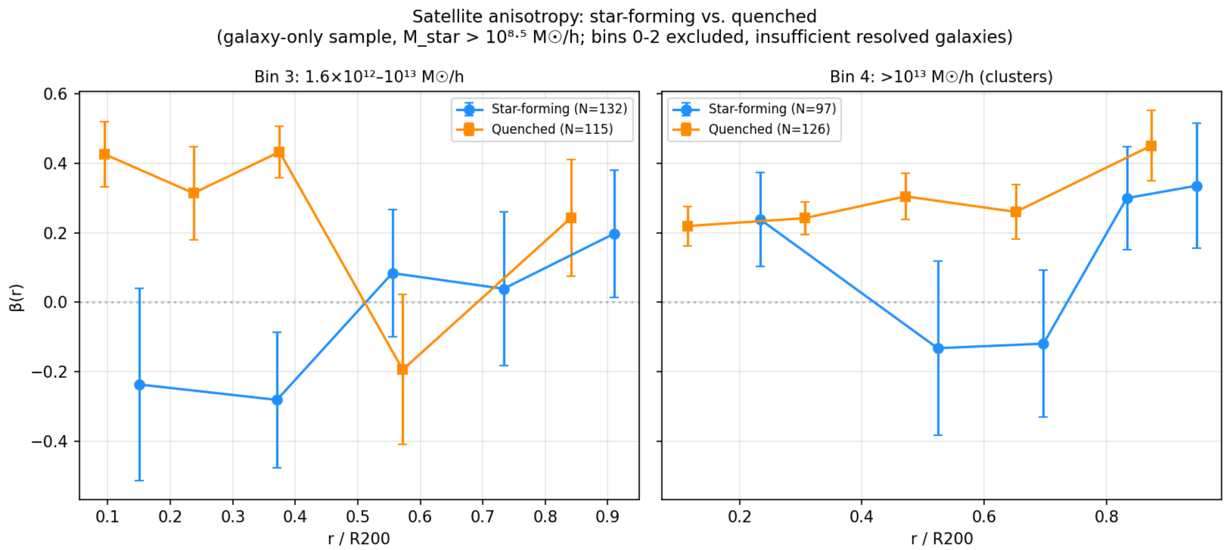

Figure 6: β split by star-forming vs. quenched satellite status, Bins 3–4.

6 Limitations

- The Bin 4 (cluster) sample is small ($N = 127$ halos), and the environment-split sign reversal in this bin should be treated as suggestive rather than confirmed.
- Bin-level Spearman correlations (across only 5 mass bins) are statistically weak; the per-halo correlation ($N = 2580$) is the defensible statistic and is what we report as significant.
- TNG100's mass resolution prevents resolving dwarf galaxies in group-mass hosts, restricting the galaxy-only cross-check to Bins 2–4.
- All results assume standard gravity, as encoded in the IllustrisTNG simulation physics.

7 Baseline for alternative dynamics

This pipeline and atlas are presented purely as a validated baseline within standard ΛCDM cosmology and standard gravity, as implemented in IllustrisTNG. No claims are made, implied, or intended regarding modified gravity, modified inertia, or any alternative dynamical framework. Future work seeking to test such frameworks against satellite kinematics would require either a simulation run under the alternative physics, or an independent observational comparison; this atlas is not designed for and should not be interpreted as evidence for or against any such framework.